

Movie Review Dataset

Problem Statements using NumPy and Pandas for data analysis and manipulation

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Dataset Overview

The dataset captures movie reviews across 10 structured columns:

Review_ID

Unique review identifier

Movie_Name / Genre

Film title and category

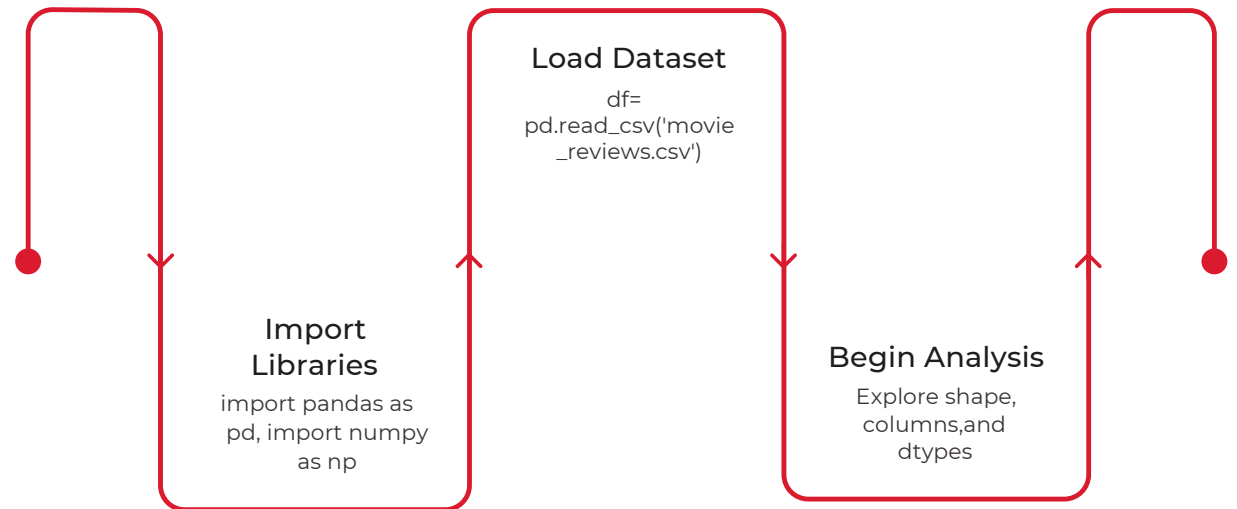
Rating / Sentiment

Score out of 10 & Positive / Negative / Neutral

Review_Length / Likes

Word count and audience engagement

Getting Started



All 20 problem statements build on this foundation 4 a single CSV loaded into a Pandas DataFrame.

Basic Statistics 4 Problems 1-4

Total Reviews

```
df.shape[0]
```

Average Rating

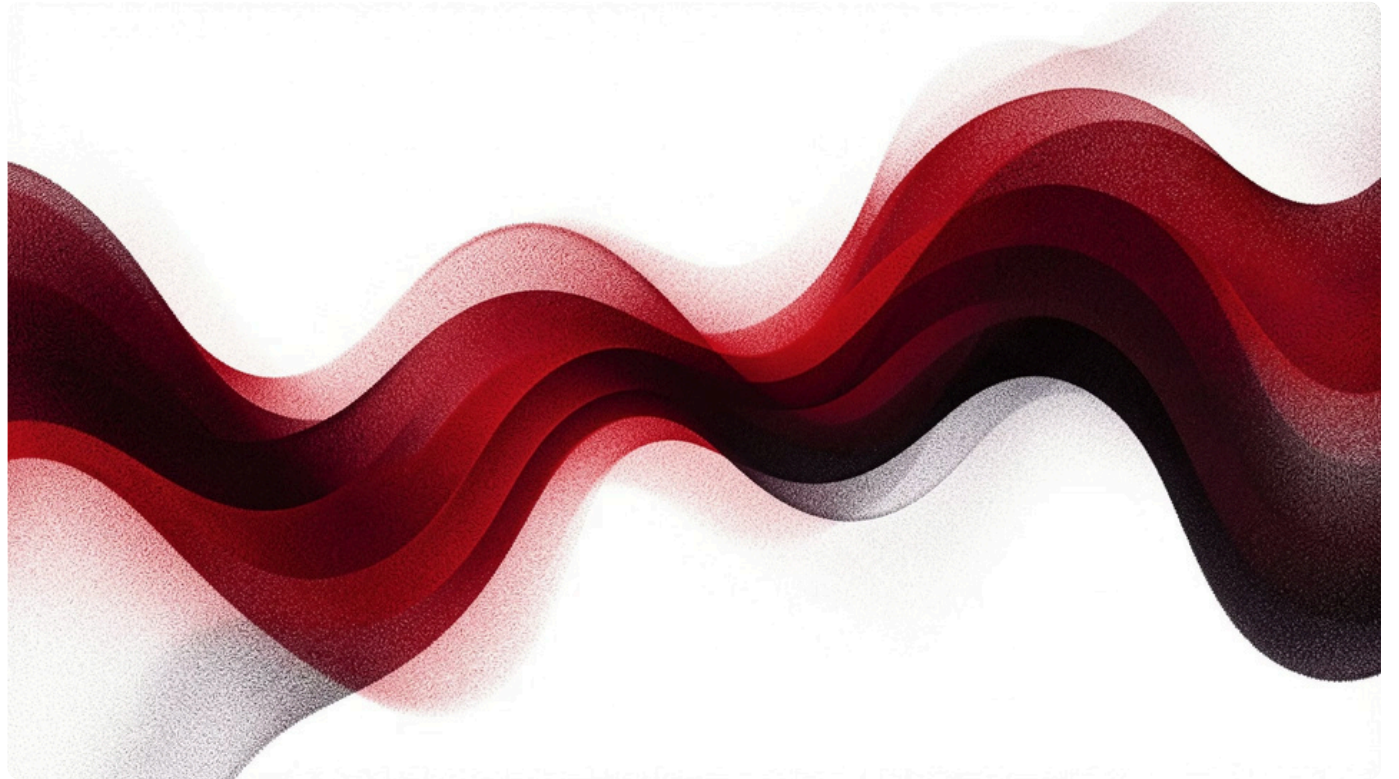
```
df[ 'Rating' ].mean( ) or  
np.mean( df[ 'Rating' ] )
```

Highest Rating

```
df.loc[df['Rating'].idxmax()]
```

Lowest Rating

```
df.loc[df['Rating'].idxmin()]
```



Sentiment & Rating Distribution :-Problems 5–8



Sentiment Count

```
df['Sentiment'].value_counts()
```

Counts Positive, Negative, and Neutral reviews.



Median Length

```
np.median(df['Review_Length'])
```

Middle value of review word counts.



High Ratings

```
df[df['Rating'] > 8]
```

Filters reviews rated above 8.



Std Deviation

```
np.std(df['Rating'])
```

Spread of ratings across all reviews.

Likes & Engagement — Problems 9–10

Total Likes

```
df['Likes'].sum()
```

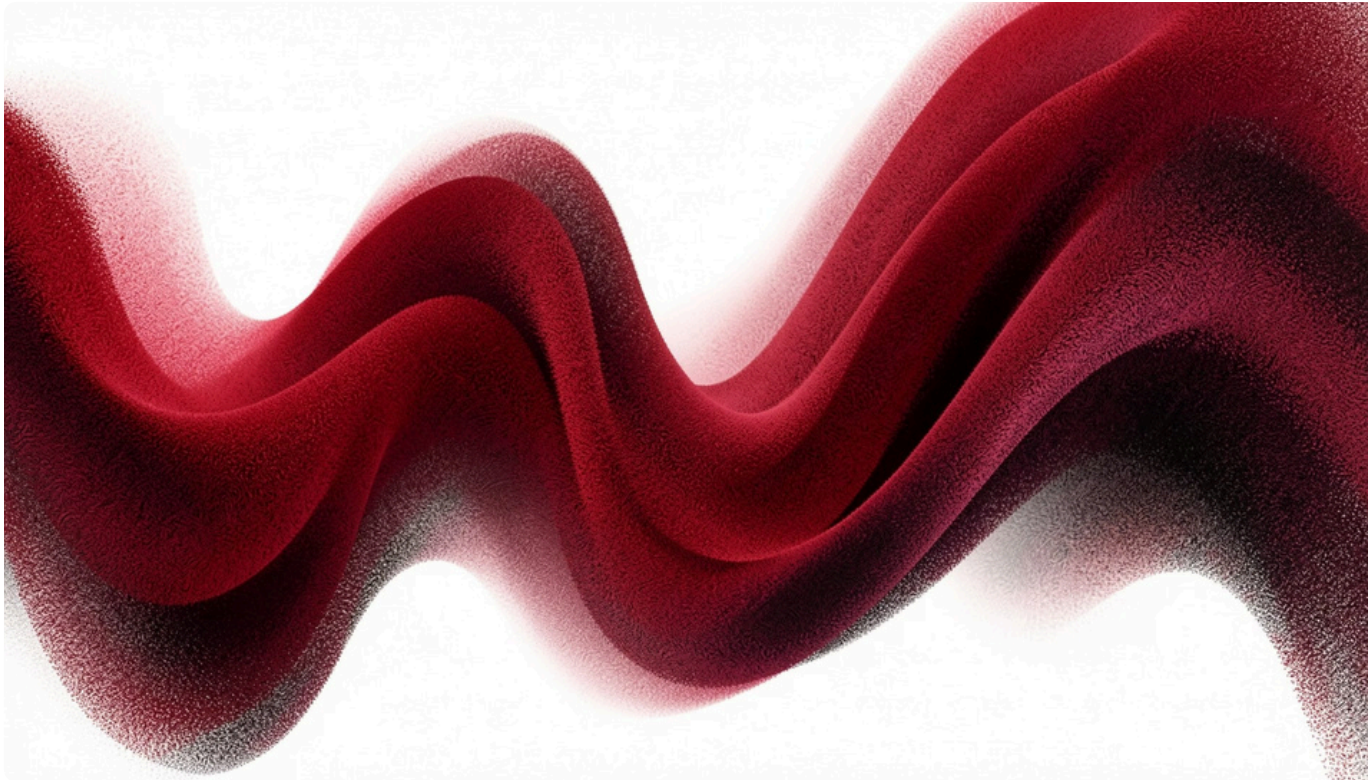
Aggregates all audience likes across every review in the dataset.

Top 5 Most Liked

```
df.nlargest(5, 'Likes')
```

Returns the five reviews with the highest like counts — a quick measure of audience resonance.

Genre Analysis 4 Problems 11-12



Average Rating by Genre

```
df.groupby('Genre')  
['Rating'].mean()
```

Groups reviews by genre and computes the mean rating for each category.

Most Reviewed Genre

```
df['Genre'].value_counts()
```

Identifies which genre dominates the dataset by review count.

Data Cleaning — Problems 13–16

01

Find Duplicates

`df[df.duplicated()]` — Flags repeated review entries.

03

Keyword Search

`df[df['Review_Text'].str.contains('excellent', case=False)]`

02

Remove Missing Values

`df.dropna()` — Drops rows with any null fields.

04

Sort by Rating

`df.sort_values(by='Rating', ascending=False)`

Advanced Analysis — Problems 17–20

Correlation: Rating vs Likes

```
df[['Rating', 'Likes']].corr()
```

NumPy Array Operations

```
ratings = np.array(df['Rating'])
```

```
Then: .max(), .min(), .mean()
```

% Positive Reviews

```
(len(df[df['Sentiment']=='Positive']) /  
len(df)) * 100
```

Monthly Review Count

```
df.groupby(df['Review_Date'].dt.month).size()
```

Conclusion

Using NumPy and Pandas , movie review datasets can be efficiently analyzed to extract insights across five key dimensions:

Ratings Analysis	Sentiment Distribution	Genre Popularity
Review Trends	Audience Engagement	